

A $\mu^{9/2} \log \mu$ UPPER BOUND FOR WINDOWED WEIL FORMS

RYOTA MORI

ABSTRACT. Let $\lambda_{\min}(\lambda)$ be the bottom of the spectrum of Weil's quadratic form restricted to test functions supported in the multiplicative window $[\lambda^{-1}, \lambda]$, and put $\mu = \lambda^2$. By Weil's criterion, the Riemann Hypothesis is equivalent to $\lambda_{\min}(\lambda) \geq 0$ for all λ . In previous papers we proved, without any hypothesis on the zeros of ζ , that $\lambda_{\min}(\lambda) \leq 1.437 \times 10^{23} \mu^{9/2} (\log \mu)^4 e^{-4\pi\mu}$ for $50 \leq \mu \leq 10^4$, using an endpoint-smoothed prolate vector. We locate the sources of the four logarithms in that bound. One logarithm is the density of the zeros of ζ at height $\asymp \mu$. The other three are artefacts. A Sobolev step weighted by the position y costs a factor $\log \mu$, which disappears after a translation by the window half-width. A sum of phase errors bounded by the triangle inequality costs $(\log \mu)^2$. Keeping the exact Prüfer phase of the prolate functions and estimating the sum in L^2 by non-stationary phase removes it. We prove, unconditionally,

$$\lambda_{\min}(\lambda) \leq 3.50 \times 10^{24} \mu^{9/2} \log \mu e^{-4\pi\mu} \quad (50 \leq \mu \leq 10^4),$$

and $\lambda_{\min}(\lambda) \leq 7.58 \times 10^{24} \mu^5 (\log \mu)^3 e^{-4\pi\mu}$ for all $\mu \geq 50$. Numerical Galerkin computations of the even part at integer μ with $7 \leq \mu \leq 20$ give ratios $\lambda_{\min}^{\text{Gal}}/(1 - \lambda_4(2\pi\mu))$ between about 4.1 and 5.4, where λ_4 is the prolate eigenvalue whose defect is of order $\mu^{9/2} e^{-4\pi\mu}$. These values are numerical, not certified. The results are upper bounds only and say nothing about positivity.

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1. INTRODUCTION

1.1. Background. Let W be Weil's quadratic form and W_λ its restriction to test functions supported in $[\lambda^{-1}, \lambda]$ [1, 2]. Write $\lambda_{\min}(\lambda)$ for the bottom of its spectrum and $\mu = \lambda^2$. By Weil's criterion, RH is equivalent to $\lambda_{\min}(\lambda) \geq 0$ for all λ . Numerically $\lambda_{\min}(\lambda)$ is positive and decays like $e^{-4\pi\mu}$.

Upper bounds come from test vectors. In [5, 6] we used the prolate vector of Connes, Consani and Moscovici [2, §7], and an endpoint-smoothed version of it, and proved

$$\begin{aligned} \lambda_{\min}(\lambda) &\leq 5.046 \times 10^{17} \mu^8 (\log \mu)^3 e^{-4\pi\mu} \quad (\mu \geq 50), \\ \lambda_{\min}(\lambda) &\leq 1.437 \times 10^{23} \mu^{9/2} (\log \mu)^4 e^{-4\pi\mu} \quad (50 \leq \mu \leq 10^4). \end{aligned}$$

The power $9/2$ is that of Fuchs' asymptotic formula for the prolate eigenvalue defect $1 - \lambda_4(c)$, $c = 2\pi\mu$ [3, 4]. Under RH and a hypothesis on close pairs of zeros, [7] proves the lower bound $\lambda_{\min} \geq \exp(-C\mu \log \log \mu)$, so even the exponent 4π is not yet matched from below, and the polynomial factor in front of $e^{-4\pi\mu}$ is open.

This paper asks how many of the four logarithms in the second bound are needed.

1.2. Where the logarithms come from. In [6, Section 6] the zeros below $T_0 = 3 \cdot 10^{12}$, where RH is verified [8], contribute

$$(1.1) \quad 48\pi [\mathfrak{A}_0 + 2\sqrt{\mathfrak{A}_0 \mathfrak{A}_1}] B, \quad \mathfrak{A}_0 = L_R Z + \frac{K_*}{R^2}, \quad \mathfrak{A}_1 = 2(a+4)^2 L_R Z + \frac{Y}{R^2},$$

with $L_R = \log(2c^3)$, $Z = Z_C(\log \mu)^2$ and $B = \rho_{\text{out}} \|h\|^2$ [6, Appendix A]. Thus the bound $\sqrt{\mathfrak{A}_0 \mathfrak{A}_1} = O((\log \mu)^4)$ is a product of three factors.

- $L_R \asymp \log \mu$ is the local density of zeros at height $\lesssim c^3$. We keep it.
- $Z \propto (\log c)^2$ comes from [5, Lemma 4.1]: $\|r_\tau\|^2 \leq \lambda^{2\tau}(1-2\tau)^{-1}(592+152.2K_1^2)B$ with $K_1 = 2\log(c/3)+8.08$. The only logarithm there is a sum $\sum_{mu \geq 2} |\rho_n(mu)|/m \leq 2\log(c/3)+5.49$. Here ρ_n is the additive residual in the representation $\tilde{\psi}_n(t) = A_{\infty,n}(ct^2)^{-1/2}[\sin(ct + \Phi_n) + \rho_n(t)]$ of [5, Lemma 3.6]. It contains the amplitude and normalisation errors and the deviation of the true phase from the linear phase $ct + \Phi_n$; the last one is not small for $t \lesssim c$.
- $(a+4)^2$, $a = \log \lambda$, comes from the weight y in the Sobolev step: $\|y q_d\|^2 \leq 2(a+4)^2 Z B$.

1.3. Results. We keep the notation of [5, 6]: $h = I_4 h_0 - I_0 h_4$, $\tilde{h}$ the smoothed vector, $\tilde{k} = \mathcal{E}(\tilde{h})|_{[\lambda^{-1}, \lambda]}$, r and $\tilde{r}$ the parts outside the window, $r_\tau(y) = e^{-\tau y} r(y)$. We continue the theorem lettering of [5, 6, 7].

Proposition 1.1 (the remainder without logarithms). *For $c \geq 10\pi$ and $0 \leq \tau < \frac{1}{2}$,*

$$\|r_\tau\|^2 \leq \frac{\lambda^{2\tau}}{1-2\tau} 729 B.$$

This replaces [5, Lemma 4.1].

Theorem J. *Without any hypothesis on the zeros of ζ :*

- (a) *for $50 \leq \mu \leq 10^4$, $|W(\tilde{k})| \leq 1.47 \times 10^{16} (\log \mu) B$;*
- (b) *for every $\mu \geq 50$, $|W(\tilde{k})| \leq 3.19 \times 10^{16} \sqrt{\mu} (\log \mu)^3 B$.*

Theorem K. *Without any hypothesis on the zeros of ζ :*

- (a) *for $50 \leq \mu \leq 10^4$, $\lambda_{\min}(\lambda) \leq 3.50 \times 10^{24} \mu^{9/2} \log \mu e^{-4\pi\mu}$;*
- (b) *for every $\mu \geq 50$, $\lambda_{\min}(\lambda) \leq 7.58 \times 10^{24} \mu^5 (\log \mu)^3 e^{-4\pi\mu}$.*

The constants are large, because they inherit the layer and trace constants of [6] (the function $V(c) \approx 1.4 \times 10^6$ below). Theorem K(a) is smaller than [6, Thm. G(a)] for all $50 \leq \mu \leq 10^4$: the ratio is $24.4/(\log \mu)^3$, which is 0.41 at $\mu = 50$ and 0.031 at $\mu = 10^4$. Theorem K(b) is smaller than [6, Thm. G(b)] exactly when $(\log \mu)^2 > 7.58 \times 10^{24}/(1.24 \times 10^{23})$, i.e. for $\mu > 2486.2$, in particular for $\mu \geq 2500$.

In the range $\mu \leq 10^4$ the restriction is where the contribution of the zeros above T_0 is absorbed into the constant, as in [6]. What is new is that the contribution of the zeros below T_0 is $O(\log \mu)B$ for every $\mu \geq 50$, and the contribution of the zeros above T_0 is $O(\sqrt{\mu}(\log \mu)^3)B$.

1.4. Is the last logarithm needed? We do not know. The remaining factor $\log \mu$ is the mean density of zeros near height c . For the unsmoothed vector of [5], Galerkin values and a heuristic computation suggest that $W(k)/(\|k\|^2(1-\lambda_4))$ grows like $\log \mu$ (Section 8); we have not computed the corresponding ratio for the smoothed vector. For $\lambda_{\min}$ itself, the numerical Galerkin ratios $\lambda_{\min}^{\text{Gal}}/(1-\lambda_4(2\pi\mu))$ at the integers of Table 1 lie between about 4.1 and 5.4, with changes of up to 22% between consecutive integers and no visible growth. This suggests:

Conjecture 1.2. $\lambda_{\min}(\lambda) \asymp 1 - \lambda_4(2\pi\mu) \asymp \mu^{9/2} e^{-4\pi\mu}$.

The upper half of Conjecture 1.2 would remove the last logarithm from Theorem K; the lower half is far beyond the methods of [7].

1.5. Method.

- (1) **Translation** (Section 6). On the real axis $|\hat{q}(t)| = |e^{iat}\hat{q}(t)|$, and $e^{iat}\hat{q}$ is the transform of $q(\cdot - a)$, which is supported in $y < 0$ (while q is supported in $y < -a$). In the Sobolev step the weight y becomes $y + a$, and the existing weighted bound with $\tau = \frac{1}{4}$ controls $\|(y + a)q\|$ by an absolute constant.
- (2) **Exact phase** (Section 3). In the Prüfer variables of [5, Lemmas 3.2 and 3.6], the exact phase satisfies $\theta' = cv(1 + \delta \sin^2 \theta)$ with $v = \sqrt{(t^2 - s)/(t^2 - 1)}$. From the bounds on δ and δ' we get $g' = (t\theta')' \geq c/2$ for $t \geq 2$.
- (3) **Orthogonality** (Section 4). The Poisson sum $\sum_m m^{-1} e^{i\theta(mu)}$ is estimated in L^2 over unit intervals in u . The phase difference of the m -th and k -th terms has derivative at least $c|m - k|/2$, and one integration by parts gives $|\int e^{i(\theta(mu) - \theta(ku))} du| \leq 10/(c|m - k|)$. Since $\sum_{m \neq k} 1/(mk|m - k|) = 4\zeta(3)$, the L^2 norm is bounded by an absolute constant.

2. SETTING AND INPUTS

Throughout, $\lambda > 1$, $\mu = \lambda^2$, $a = \log \lambda$, $c = 2\pi\mu \geq 10\pi$, and $n \in \{0, 4\}$ indexes the two prolate functions. Fourier transforms are $\hat{f}(z) = \int f(y) e^{izy} dy$ in the logarithmic variable. For a function f on $(-\infty, -a)$ put $f_\tau(y) = e^{-\tau y} f(y)$. The quantities $\psi_n, \tilde{\psi}_n, \lambda_n, \chi(n), w_n, \Psi, \alpha = I_4, \beta = -I_0, B = \rho_{\text{out}} \|h\|^2$ are those of [5, Section 2].

From [5] we use:

- the Poisson form [5, Lemma 2.3]: $r(-\log X) = X^{1/2} \sum_{m \geq 1} \hat{h}(mX) - \frac{1}{2} h(0) X^{-1/2}$ for a.e. $X > \lambda$, with symmetric partial sums; and $\hat{h}(\lambda t) = \sum_n w_n \tilde{\psi}_n(t)$;
- the Liouville–Green normal form [5, Lemma 3.2]: with $p = t^2 - 1$, $r_s = t^2 - s$, $q = c^2 r_s$, $s = \chi_n^{\text{SL}}/c^2 \in [0, 0.3)$, $z' = \sqrt{q/p}$ and $\tilde{\psi}_n = (pq)^{-1/4} w$, one has $w_{zz} + (1 + \delta)w = 0$ with

$$\delta = \frac{1}{4c^2} \left[\frac{1}{p r_s} - \frac{s^2 + 2s + (3 - 6s)t^2}{r_s^3} \right], \quad |\delta| \leq \frac{0.5038}{c^2 t^4} \quad (t \geq 2);$$

- the amplitude bounds [5, Lemma 3.5]: $A_{\infty, n} \leq 1.78 |\psi_n(1)|$ for the limit $A_{\infty, n}$ of $(w^2 + w_z^2)^{1/2}$, and $\tilde{\psi}_n(t)^2 \leq 3.79 \psi_n(1)^2 / (c\sqrt{t^2 - 1})$ for $1 < t \leq 2$;
- from the proof of [5, Lemma 3.6], for $t \geq 2$: the amplitude $A(t) = (w^2 + w_z^2)^{1/2}$ satisfies $|A(t)/A_\infty - 1| \leq 0.00464 t^{-2}$, and $(pq)^{-1/4} = (ct^2)^{-1/2} (1 + \eta)$ with $0 \leq \eta \leq 0.3829 t^{-2}$;
- $\Psi^2 \leq 8cB$ [5, Lemma 3.3] and $h(0)^2 \leq 2.0408 \lambda B$ [5, Lemma 3.4].

From [6] we use the smoothed vector, the trace J and corrector b_d , the function $V(c)$ with $\text{TV}(\tilde{h} - h) \leq V(c)\sqrt{\lambda B}$, condition (A) and the zero-side machinery of [6, Sections 2–6 and Appendix A]. The zero count is that of [9, Cor. 1] with the constants 0.112, 0.278, 2.510 of its published version, as in version 2 of [5, 6].

3. THE EXACT PHASE

Fix $n \in \{0, 4\}$. Since (w, w_z) never vanishes for a nonzero solution, we may write $w = A \sin \theta$, $w_z = A \cos \theta$ with $A > 0$ and a continuous phase θ . Differentiating gives

$$(3.1) \quad \theta_z = 1 + \delta \sin^2 \theta, \quad (\log A)_z = -\delta \sin \theta \cos \theta.$$

We write $\Theta(t) = \theta(z(t))$ and use $'$ for d/dt . Then, with $v = \sqrt{(t^2 - s)/(t^2 - 1)}$,

$$(3.2) \quad \Theta' = cv(1 + \delta \sin^2 \Theta).$$

Lemma 3.1 (far-field representation with the exact phase). *For $t \geq 2$,*

$$\tilde{\psi}_n(t) = \frac{A_{\infty, n}}{\sqrt{c}t} [\sin \Theta_n(t) + r_n^{\text{amp}}(t)], \quad |r_n^{\text{amp}}(t)| \leq \frac{0.4}{t^2}.$$

Proof. $\tilde{\psi}_n = (pq)^{-1/4} A \sin \Theta = (ct^2)^{-1/2} (1 + \eta)(1 + \varepsilon_2) A_\infty \sin \Theta$ with $|\varepsilon_2| \leq 0.00464t^{-2}$ and $0 \leq \eta \leq 0.3829t^{-2}$. Hence $r^{\text{amp}} = (\varepsilon_2 + \eta + \varepsilon_2\eta) \sin \Theta$, and $0.00464 + 0.3829 + 0.00464 \cdot 0.3829/4 < 0.4$ since $t \geq 2$. No phase approximation is used. $\square$

Lemma 3.2 (derivatives). *For $t \geq 2$ and $c \geq 10\pi$, put $q_\theta = \Theta' - cv$ and $g(t) = t\Theta'(t)$. Then*

$$|q_\theta| \leq \frac{6}{5ct^4}, \quad |q'_\theta| \leq \frac{15}{ct^5} + \frac{3}{2t^4}, \quad g'(t) \geq \frac{c}{2}.$$

Proof. For $t \geq 2$ we have $1 \leq v \leq \sqrt{4/3} < 6/5$ and $v' = -(1-s)t/[(t^2-1)^2v]$, so $|v'| \leq 2/t^3$. We weaken $|\delta| \leq 0.5038/(c^2t^4)$ to $1/(c^2t^4)$. Then (3.2) gives $|q_\theta| \leq 6/(5ct^4)$ and $\Theta' \leq 5c/4$.

For δ' , write $\delta = (4c^2)^{-1}[1/(pr_s) - N/r_s^3]$ with $N = s^2 + 2s + (3-6s)t^2$. For $t \geq 2$ and $0 \leq s < 0.3$ we have $p \geq \frac{3}{4}t^2$, $r_s \geq \frac{37}{40}t^2$, $0 \leq N \leq (3+0.69/4)t^2$ and $|N'| \leq 6t$. Differentiating the three factors,

$$|\delta'| \leq \frac{1}{4c^2t^5} \left[\frac{4}{(3/4)^2(37/40)^2} + \frac{6}{(37/40)^3} + \frac{6(3+0.69/4)}{(37/40)^4} \right] < \frac{12}{c^2t^5};$$

the bracket divided by 4 equals $176656000/16867449 < 10.48$. Now

$$q'_\theta = cv'\delta \sin^2 \Theta + cv\delta' \sin^2 \Theta + cv\delta \sin(2\Theta)\Theta'.$$

The three terms are at most $2/(ct^7)$, $14.4/(ct^5)$ and $\frac{6}{5} \cdot \frac{5}{4} t^{-4} = \frac{3}{2}t^{-4}$. The last term comes from the fast oscillation of Θ and is kept. Since $t \geq 2$, the first two add up to less than $15/(ct^5)$.

Let $h_1(t) = tv(t)$. Then $h'_1 = v[1 - (1-s)t^2/((t^2-1)(t^2-s))]$, and $(1-s)t^2/((t^2-1)(t^2-s)) \leq t^2/(t^2-1)^2 \leq 4/9$, so $\frac{5}{9} \leq h'_1 \leq \frac{6}{5}$. Since $g = ch_1 + tq_\theta$,

$$|g' - ch'_1| \leq |q_\theta| + t|q'_\theta| \leq \frac{17}{ct^4} + \frac{3}{2t^3} \leq \frac{17}{480} + \frac{3}{16} < \frac{c}{18},$$

and $g' \geq \frac{5}{9}c - \frac{1}{18}c = \frac{c}{2}$. $\square$

4. ORTHOGONALITY OF THE POISSON TERMS

Let $J_j = [j, j+1]$, $j \geq 1$. On J_1 we use $m \geq 2$ and on J_j , $j \geq 2$, we use $m \geq 1$; thus $mu \geq 2$ for every term.

Lemma 4.1 (one pair). *Let $m > k$ be admissible indices on J_j , $D = m - k$, and $\phi(u) = \Theta(mu) - \Theta(ku)$. Then*

$$\left| \int_{J_j} e^{i\phi(u)} du \right| \leq \frac{10}{cD}.$$

Proof. $\phi'(u) = [g(mu) - g(ku)]/u \geq cD/2$ by Lemma 3.2. Let $\zeta' = cv$ be the WKB phase and $\phi_0(u) = \zeta(mu) - \zeta(ku)$. Since $h'_1 \leq \frac{6}{5}$, $0 \leq \phi'_0 \leq \frac{6}{5}cD$. The function $t^2v'(t) = -(1-s)t^3/[(t^2-1)^{3/2}(t^2-s)^{1/2}]$ is negative and increasing, since the logarithmic derivative of its absolute value is $3/t - 3t/(t^2-1) - t/(t^2-s) < 0$. Hence $\phi''_0 = (c/u^2)[(mu)^2v'(mu) - (ku)^2v'(ku)] \geq 0$ and $\int_{J_j} |\phi''_0| \leq \phi'_0(j+1) \leq \frac{6}{5}cD$. By Lemma 3.2,

$$|\phi'' - \phi''_0| \leq \frac{15}{cu^5}(m^{-3} + k^{-3}) + \frac{3}{2u^4}(m^{-2} + k^{-2}),$$

whose integral over J_j is at most $30/c + 3 \leq 4$. Integrating $e^{i\phi} = (e^{i\phi})'/(i\phi')$ by parts,

$$\left| \int_{J_j} e^{i\phi} \right| \leq \frac{2}{\min \phi'} + \int_{J_j} \frac{|\phi''|}{\phi'^2} \leq \frac{4}{cD} + \frac{4}{c^2D^2} \left(\frac{6}{5}cD + 4 \right) \leq \frac{28/3}{cD}.$$

$\square$

Lemma 4.2 (the Poisson sum in L^2). *Let $S(u) = \sum_m m^{-1}e^{i\Theta(mu)}$, the sum over the admissible m . The partial sums converge in $L^2(J_j)$, uniformly in j , and $\|S\|_{L^2(J_j)}^2 \leq \zeta(2) + 40\zeta(3)/c < (11/6)^2$.*

Proof. Expanding the square, the diagonal contributes at most $\zeta(2)$ and, by Lemma 4.1, the off-diagonal terms at most $(10/c) \sum_{m \neq k} 1/(mk|m-k|)$. Writing $d = |m-k|$, $\sum_{m \neq k} 1/(mk|m-k|) = 2 \sum_{m,d \geq 1} 1/(md(m+d)) = 2 \int_0^1 \log^2(1-x) dx/x = 4\zeta(3)$. The same estimate restricted to $m, k > N$ tends to 0 uniformly in j , so the partial sums are Cauchy in $L^2(J_j)$. Finally $\zeta(2) + 40\zeta(3)/(10\pi) < 10/3 < (11/6)^2$. $\square$

5. THE REMAINDER: PROOF OF PROPOSITION 1.1

Put $X = \lambda u$ with $u > 1$, and split $r(-\log X)$ into three parts: *far* (all terms with $mu \geq 2$), *edge* ($m = 1, 1 < u < 2$) and *central* ($-\frac{1}{2}h(0)X^{-1/2}$). The parts cover $u > 1$ without overlap. The measure is $dy = du/u$ and $e^{-\tau y} = (\lambda u)^\tau$, so for any F ,

$$\|F_\tau\|^2 = \lambda^{2\tau} \int_1^\infty u^{2\tau-1} |F|^2 du.$$

Far part. By the Poisson form and Lemma 3.1, with $b_n = \lambda^{1/2} w_n A_{\infty,n} c^{-1/2}$,

$$r_{\text{far}}(-\log \lambda u) = u^{-1/2} \sum_n b_n \sum_{mu \geq 2} \frac{\sin \Theta_n(mu) + r_n^{\text{amp}}(mu)}{m}.$$

By Lemma 4.2, the sine sum has $L^2(J_j)$ norm at most $11/6$. The amplitude remainders sum absolutely: $\sum_m |r_n^{\text{amp}}(mu)|/m \leq 0.4\zeta(3)/u^2 < \frac{1}{2}$. We combine $n = 0$ and $n = 4$ by Minkowski, so no separation between the two phases is needed. With $\sum_n |b_n| \leq 1.78\Psi/\sqrt{c} \leq 1.78\sqrt{8B}$ and $\sum_{j \geq 1} j^{2\tau-2} \leq 1 + 1/(1-2\tau) \leq 2/(1-2\tau)$,

$$\|r_{\text{far},\tau}\| \leq \frac{\lambda^\tau \sqrt{B}}{\sqrt{1-2\tau}} \cdot 4 \cdot 1.78 \cdot \frac{7}{3}.$$

We first apply this to finite symmetric partial sums. By Lemma 4.2 they converge in the weighted L^2 norm, and a subsequence converges a.e. to the function given by the Poisson form.

Edge part. Here $X^{1/2}\hat{h}(X) = (\lambda u)^{1/2} \sum_n w_n \hat{\psi}_n(u)$ and $\sum_n |w_n| |\psi_n(1)| = \Psi/\sqrt{\lambda}$, so $|r_{\text{edge}}|^2 \leq 3.79 u \Psi^2 / (c\sqrt{u^2-1})$. Since $u^{2\tau} \leq 2$ on $(1, 2)$,

$$\|r_{\text{edge},\tau}\|^2 \leq \lambda^{2\tau} \frac{7.58\Psi^2}{c} \operatorname{arcosh} 2 \leq \lambda^{2\tau} B \cdot 8 \cdot 7.58 \operatorname{arcosh} 2 < \frac{\lambda^{2\tau}}{1-2\tau} 81B.$$

Central part. $\|r_{\text{central},\tau}\|^2 = \lambda^{2\tau} h(0)^2 / (4\lambda(1-2\tau)) \leq \lambda^{2\tau} B \cdot 0.5102 / (1-2\tau) < \lambda^{2\tau} (0.72)^2 B / (1-2\tau)$.

By Minkowski, $\|r_\tau\| \leq (16.62 + 9 + 0.72)\lambda^\tau \sqrt{B/(1-2\tau)} < 27\lambda^\tau \sqrt{B/(1-2\tau)}$, which proves Proposition 1.1. For $-\frac{1}{2} < \tau < 0$ we have $\|r_\tau\| \leq \lambda^\tau \|r_0\|$ and $1-2\tau \leq 2$, so for all $|\tau| < \frac{1}{2}$,

$$(5.1) \quad \|r_\tau\|^2 \leq \frac{\lambda^{2\tau}}{1-2\tau} 1458 B.$$

$\square$

6. TRANSLATION IN THE SOBOLEV STEP

Let $q = q_d$ be the corrected remainder of [6, Section 5], and let Z, K be numbers with

$$\|q_\tau\|^2 \leq \frac{\lambda^{2\tau}}{1-2\tau} ZB, \quad \|(q_\tau)'\|^2 \leq \frac{\lambda^{2\tau}}{1-2\tau} KB \quad (|\tau| < \tfrac{1}{2}).$$

Lemma 6.1 (translation). $\|(y+a)q\|^2 \leq (32/e^2)ZB$ and $\|((y+a)q)'\|^2 \leq Y_{\text{tr}}B$ with $Y_{\text{tr}} = [\sqrt{Z} + (\sqrt{32}/e)(\sqrt{K} + \frac{1}{4}\sqrt{Z})]^2$.

Proof. Put $\sigma = -y - a \geq 0$ on the support of q . With $\tau = \frac{1}{4}$, $e^{-y/2} = \lambda^{1/2} e^{\sigma/2}$, so $\int e^{\sigma/2} |q|^2 \leq 2ZB$. Since $\sup_{\sigma \geq 0} \sigma^2 e^{-\sigma/2} = 16/e^2$, the first claim follows. With $w = e^{-y/4} q$ we have $e^{-y/4} q' = w' + w/4$, hence $\int e^{\sigma/2} |q'|^2 \leq 2(\sqrt{K} + \frac{1}{4}\sqrt{Z})^2 B$. Apply Minkowski to $((y+a)q)' = q + (y+a)q'$. The zero extension of q is in H^1 with zero trace at $y = -a$, so there are no boundary terms. $\square$

On the real axis, $G(t) = e^{iat} \hat{q}(t)$ satisfies $|G| = |\hat{q}|$ and $G' = ie^{iat} (\widehat{y+a} q)$. We apply the Sobolev bound [6, (6.1)] to G instead of $\hat{q}$. We first split $|F|^2 \leq 2|\hat{q}|^2 + 2|\hat{b}_d|^2$ as in [6]; the b_d part is unchanged. The translation is used only on the real axis, i.e. for the zeros below T_0 , which lie on the critical line [8]. For the zeros above T_0 we use the disc averages of [6, Section 6] unchanged, with their factors $\lambda^{2\tau}$. With $\log(|t| + e + 1) \leq L_R + t^2/R^2$, $R = c^3$, and Plancherel, the low zeros contribute at most

$$(6.1) \quad 48\pi [A_0 + 2\sqrt{A_0 A_1}] B + 4D_+(c) S(c^2) B, \quad A_0 = L_R Z + \frac{K}{R^2}, \quad A_1 = \frac{32}{e^2} L_R Z + \frac{Y_{\text{tr}}}{R^2},$$

with D_+ and S as in [6, Appendix A].

7. ASSEMBLY

New Z and K . In [6, (3.3)], $m_0(c)$ is built from [5, Lemma 4.1] and $V(c)$. With (5.1) it becomes

$$m_{\text{new}}(c) = [\sqrt{1458} + V(c)]^2, \quad Z_{\text{new}}(c) = [\sqrt{m_{\text{new}}(c)} + \sqrt{D_J(c)/c}]^2.$$

Both decrease in c , because V decreases and $\sqrt{D_J}/c = \sqrt{8/c} + E_J(c)/c$ with $E_J = O(\log c)$. At $c = 100\pi$, ball arithmetic gives $m_{\text{new}} < 1.906 \times 10^{12}$ and $Z_{\text{new}} < 1.918 \times 10^{12}$, so $\bar{m} = \bar{Z} = 2 \times 10^{12}$ serve for all $\mu \geq 50$. The logarithm of E_J is not removed; it is divided by c . Rerunning condition (A) [6, Prop. 5.3] with $\bar{Z}$ gives $K_{\text{new}} = [\sqrt{\bar{Z}} + g_A c^2 + 120c^{3/2} + c\sqrt{D_+(c)}]^2$ and, by Lemma 6.1, $Y_{\text{new}} = [\sqrt{\bar{Z}} + (\sqrt{32}/e)(\sqrt{K_{\text{new}}} + \frac{1}{4}\sqrt{\bar{Z}})]^2$.

Majorants. Let $c_* = 100\pi$ and $l = \log \mu \geq l_* = \log 50$. Since $D_+(c)/c^2$ decreases, $K_{\text{new}} \leq \bar{K} c^4$ and $Y_{\text{new}} \leq \bar{Y} c^4$, with $\bar{K} = [\sqrt{\bar{Z}}/c_*^2 + g_A + 120/\sqrt{c_*} + \sqrt{D_+(c_*)}/c_*]^2$ and $\bar{Y} = [\sqrt{\bar{Z}}/c_*^2 + (\sqrt{32}/e)(\sqrt{\bar{K}} + \sqrt{\bar{Z}}/(4c_*^2))]^2$. Since $L_R = \log 2 + 3 \log c \leq 5l$ and $c^2 l$ increases, $A_0 \leq a_0 l$ and $A_1 \leq a_1 l$ with $a_0 = 5\bar{Z} + \bar{K}/(c_*^2 l_*)$ and $a_1 = 5(32/e^2)\bar{Z} + \bar{Y}/(c_*^2 l_*)$. The term $D_+(c)S(c^2)$ is a sum of terms $(\log c)^p/c^q$ with positive coefficients and $p/q \leq \frac{3}{2} < \log c_*$, so it decreases. Hence the low zeros contribute at most $C_L l B$ with

$$C_L = 48\pi [a_0 + 2\sqrt{a_0 a_1}] + 4D_+(c_*) S(c_*^2)/l_* < 7.785 \times 10^{15}.$$

Proof of Theorem J(a). Take $T = T_0$ and the strip bound $L_T(\lambda)$ of [6] for the zeros above T_0 . Using $K_{\text{new}} \leq \bar{K}(2\pi \cdot 10^4)^4$, $L_T(\lambda) \leq L_T(100)$, $\lambda \leq 100$, the monotonicity of D_+ and $l \geq l_*$,

$$\frac{|W(\tilde{k})|}{lB} \leq C_L + \frac{2A(T_0)L_T(100)\bar{K}(2\pi \cdot 10^4)^4 + 400D_+(2\pi \cdot 10^4)(\log T_0 + 1)/(\pi T_0)}{l_*} < 1.47 \times 10^{16}.$$

Of this constant, about 6.8×10^{15} comes from the zeros above T_0 ; it is not negligible.

Proof of Theorem J(b). Take $T = \max(T_0, c^3)$ and $\mu_* = T_0^{1/3}/(2\pi)$. The intermediate zeros $T_0 < |t_\rho| \leq T$ (only for $\mu > \mu_*$) contribute $2n(x)(11.15x)^2 \lambda j(x, l) \bar{m} B$ with $x = \log(3c^3)$ [6, Section 6], and the zeros above T contribute as in [6] with K_{new} . Since x_T/l and $n(x)/x$ decrease and j increases in x/l , and with $T^2 \geq T_0^{2/3} c^4$, $T \geq c^3$ and $\log T + 1 \leq x_T$,

$$|W(\tilde{k})| \leq [C_L l + C_M \lambda^3 \mathbf{1}_{\mu > \mu_*} + C_T \lambda^3 + C_B \lambda] B,$$

with $C_M < 3.175 \times 10^{16}$, $C_T < 1.165 \times 10^{10}$ and $C_B < 3.493 \times 10^6$. Dividing by λ^3 and using $\mu \geq 50$ gives the constant 3.19×10^{16} .

Proof of Theorem K. As in [6, Section 7], $\lambda_{\min} \leq |W(\tilde{k})|/(0.1089\|h\|^2)$ and $B \leq 6621c^{9/2}e^{-2c}\|h\|^2$ [4], and $6621(2\pi)^{9/2}/0.1089 \leq 237522697$. Then $237522697 \times 1.47 \times 10^{16} < 3.50 \times 10^{24}$ and $237522697 \times 3.19 \times 10^{16} < 7.58 \times 10^{24}$. $\square$

8. NUMERICAL CONTEXT

Galerkin values. The bottom of the even part of W_λ was computed on the Legendre basis of $L^2(-a, a)$, as in [5, Section 9], with N basis functions. The matrix entries are computed in ball arithmetic, but the quadrature errors are not enclosed, so these values are numerical. Rayleigh–Ritz values computed from the exact matrix would be upper bounds for the bottom of the even part, and hence for $\lambda_{\min}$; since the quadrature errors are not enclosed, the values below are not certified upper bounds. Table 1 lists the ratio to the prolate defect $1 - \lambda_4(2\pi\mu)$.

μ	7	11	12	13	14	15	16	17	18	19	20
N	240	320	300	330	370	370	440	420	440	470	570
ratio	4.21	4.72	4.11	4.94	4.55	4.63	4.49	4.70	4.38	5.34	4.90

TABLE 1. $\lambda_{\min}^{\text{Gal}}/(1 - \lambda_4(2\pi\mu))$ at the largest computed N . At $\mu = 18, 19$ only one N was computed. At $\mu = 20$ the values for $N = 350, 390, 440, 500, 570$ are 7.83, 5.87, 5.30, 5.02, 4.90; the last one was computed with 1300 bits.

Three observations:

- *Resolution.* In these computations, convergence in N needed $N/(ac) \gtrsim 2.3$ rather than a fixed N/c ; this is an empirical guide, not a sufficient condition. With too small N the ratio is badly overestimated: for example 7.83 at $\mu = 20$, $N = 350$, and 38.4 at $\mu = 25$, $N = 440$. An earlier impression that the ratio grows from 3.6 to 5.4 on $5 \leq \mu \leq 17$ came from such under-resolved values.
- *Accuracy.* Recomputing with more precision (980 instead of 680 bits at $\mu = 17$, $N = 420$; 1100 instead of 760 bits at $\mu = 20$, $N = 440$) and 100 more quadrature nodes reproduced the stored 8 significant digits of $\lambda_{\min}^{\text{Gal}}$ (relative differences 1.5×10^{-8} and 7.6×10^{-9}). The required precision grows with N : at $\mu = 20$, $N = 570$, a run with 840 bits gave a negative bottom, while 1300 bits (and 100 more nodes) gave $1.09899 \times 10^{-96} > 0$, the value in Table 1. Precision was checked only at these four points; at $\mu = 25$ ($N \leq 660$, at most 1000 bits) the values were still decreasing (5.86 at $N = 580$, 5.14 at $N = 660$) and were not checked, so we do not list them.
- *Fluctuation.* The ratio changes by up to 22% between consecutive integers (for example $4.11 \rightarrow 4.94$ from $\mu = 12$ to 13, and $4.38 \rightarrow 5.34$ from 18 to 19), while $1 - \lambda_4(2\pi\mu)$ is smooth: its ratio to Fuchs’ asymptotic $\frac{2^{11}\sqrt{\pi}}{3}c^{9/2}e^{-2c}$ increases smoothly from 0.89 at $\mu = 11$ to 0.94 at $\mu = 20$. A possible explanation is arithmetic: the prime-power terms $n < \mu$ of the explicit formula enter one at a time as μ grows. We have not tested this.

The test vector. For the unsmoothed vector of [5], the Galerkin values of $W(k)/(\|k\|^2(1 - \lambda_4))$ are 5.4, 6.0, 6.8, 7.0, 7.5, 7.4 at $\mu = 5, 7, 11, 13, 15, 17$. This is consistent with growth like $A \log \mu + B$. Heuristically, under RH and counting zeros with multiplicity, $W(k) = 2 \sum_{\gamma > 0} m_\gamma |\hat{k}(\gamma)|^2$, and $|\hat{k}(t)|^2$ is roughly flat up to $t \asymp c$ and decays like t^{-2} beyond. Summing this against the mean zero density $\frac{1}{2\pi} \log \frac{t}{2\pi}$ (the derivative of the main term of the zero count) gives a factor $\log \mu$. We have not proved this. Note that this mean density equals the Nyquist density a/π of the window exactly at height $t = c = 2\pi\mu$, which links the prolate bandwidth of [5] to the sampling picture of [7].

REPRODUCIBILITY

The constants of Sections 3–7 are computed with Arb (python-flint) at 256 bits. Every printed upper bound of Sections 7 and 1.3 is asserted against the ball. The scripts (proofs-V/constants.V.py), their outputs, and the Galerkin scripts and raw outputs of Section 8

(`numerics-V/`) are in release `v1.3.0` of the public repository <https://github.com/moriryota/prolate-weil-bounds>, archived on Zenodo, doi:10.5281/zenodo.23092422 (all versions).

USE OF GENERATIVE AI

This work was carried out with extensive use of generative AI systems, namely Anthropic Claude and OpenAI GPT. Under the direction of the author, these systems located the logarithmic losses, drafted the proofs, implemented and ran the computations, and reviewed one another's work adversarially. Two independent AI reviews of the new arguments found no blocking or major issues, and an AI citation audit checked the versions and constants of the cited results. The author is responsible for the content.

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INDEPENDENT RESEARCHER, TOKYO, JAPAN
 Email address: moriryota31@gmail.com